SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct

I Shaik Ashraf/192324088 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning.
Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Question 1: 1. Media Streaming Platform

Sol:

Scenario Understanding: Millions of users should stream videos with very low buffering.

Challenges: High traffic, latency, storage demand and sudden traffic spikes.

Proposed Cloud Architecture: Use CDN + Load Balancer + Auto Scaling + Object Storage + Media Servers + Database.

Load Balancing & Auto Scaling: Load balancer distributes requests. Auto Scaling adds/removes servers automatically during peak hours.

Cost Optimization: Use reserved instances for base load, spot instances for extra capacity, lifecycle policies and CDN caching.

Conclusion: Architecture provides high availability, scalability and lower operational cost.

Technique	Purpose
CDN	Deliver content from nearest location
Load Balancer	Distribute user requests
Auto Scaling	Increase or decrease servers
Object Storage	Store videos reliably

Question 2: 2. FinTech Multi-cloud Containers

Sol:

Scenario: Applications run across multiple cloud providers.

Issues: Deployment inconsistency and difficult management.

Solution: Use Kubernetes with container registry and Infrastructure as Code.

Benefits: Service discovery through Kubernetes DNS; automatic scaling using Horizontal Pod Autoscaler.

Risks: Complex management, higher cost and security policy differences.

Conclusion: Kubernetes ensures consistent deployment across clouds.

Benefit	Description
High Availability	Workloads continue if one cloud fails
Vendor Independence	Avoid vendor lock-in
Scalability	Resources increase automatically

Question 3: 3. Backup & Disaster Recovery

Sol:

Scenario: Data is lost due to accidental deletion.

Risks: Permanent data loss and downtime.

Solution: Enable automated backups, replication, snapshots and versioning.

RTO & RPO: Frequent backups reduce RPO; fast recovery and standby systems reduce RTO.

Business Continuity: Regular testing, monitoring and disaster recovery drills.

Conclusion: Cloud-native backup ensures reliable recovery.

Component	Purpose
Snapshot	Quick recovery
Replication	Copies data to another region
Versioning	Restore deleted files

Question 4: 4. Hybrid Cloud

Sol:

Scenario: Sensitive data stays on-premises while other workloads run in cloud.

Architecture: Connect private datacenter and cloud using VPN or dedicated connection.

Identity: Implement Single Sign-On, IAM and Multi-Factor Authentication.

Encryption: Encrypt data during storage and transmission.

Challenges: Network latency, compliance and management complexity.

Conclusion: Hybrid cloud balances security with flexibility.

Data Type	Deployment
Sensitive	On-Premises
Non-sensitive	Public Cloud

Question 5: 5. Multi-region High Availability

Sol:

Scenario: Application must continue even if one region fails.

Solution: Deploy application in two or more regions with global load balancer.

Replication: Use synchronous/asynchronous replication and automatic failover.

Monitoring: Monitor using cloud monitoring dashboards, alerts and logs.

Benefits: High availability, fault tolerance and better disaster recovery.

Conclusion: Multi-region deployment improves reliability.

Mechanism	Purpose
Global Load Balancer	Route traffic
Failover	Switch to healthy region
Monitoring	Detect issues early